Security Audit Report

Authorized password-strength assessment & storage-security analysis

AUTHORIZED USE ONLY. This report was produced by an authorized password security assessment against SYNTHETIC test data. It must not be used to access accounts without explicit permission. No real user secrets are collected or stored.

Audit Overview

Dataset	Secure store PBKDF2 (salted)
Attack type	all
Started	2026-11-02T13:04:33+00:00
Finished	2026-11-02T13:04:33+00:00
Duration (s)	16.760
Accounts audited	17
Algorithms	pbkdf2_sha256

Results Summary

Passwords recovered: 3 / 17 (17.6%)

Overall risk: Critical Storage grade: Good

Total candidate attempts: 17

Risk breakdown - Critical: 3 High: 0 Medium: 0 Low: 14

Password Storage Assessment

- Passwords use salted, slow hashing -- resistant to precomputation and batch attacks. Weak passwords can still fall to targeted guessing.

Per-Account Findings

Username	Algo	Cracked	Method	Risk	Recovered
alice	pbkdf2_sha25	Yes	dictionary	Critical	password
bob	pbkdf2_sha25	Yes	dictionary	Critical	123456
carol	pbkdf2_sha25	Yes	dictionary	Critical	qwerty
dave	pbkdf2_sha25	No	-	Low	-
erin	pbkdf2_sha25	No	-	Low	-
frank	pbkdf2_sha25	No	-	Low	-
grace	pbkdf2_sha25	No	-	Low	-
heidi	pbkdf2_sha25	No	-	Low	-
ivan	pbkdf2_sha25	No	-	Low	-
judy	pbkdf2_sha25	No	-	Low	-
mallory	pbkdf2_sha25	No	-	Low	-
niaj	pbkdf2_sha25	No	-	Low	-
olivia	pbkdf2_sha25	No	-	Low	-
peggy	pbkdf2_sha25	No	-	Low	-
rupert	pbkdf2_sha25	No	-	Low	-
sybil	pbkdf2_sha25	No	-	Low	-
trent	pbkdf2_sha25	No	-	Low	-

Recommendations

- 3 of 17 passwords were recovered (17.6%). Force resets for all affected accounts.
- Enforce a minimum length of 12+ characters and screen new passwords against a common-password blocklist.
- Enable multi-factor authentication to reduce the impact of any single compromised password.

Method Limitations

- Dictionary: only finds passwords present in the wordlist.
- Hybrid: only rule-expressible mutations; candidate space grows fast.
- Rainbow/lookup: storage grows with coverage and is defeated by salting; this tool uses a direct hash-lookup table, not reduction-chain rainbow tables.
- Results are bounded by the chosen wordlist and single-machine Python speed; a dedicated GPU cracker would be faster.